



Lokacija	DAFRA D.O.O.
Datum	28.10.2024 – 30.10.2024

Linearizacija sile	$F = 11,71 \cdot l - 3,09 \text{ (N)}$
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Testiranje sile (vertikalna sila na pestiču) Vzmet (0,55mmx4mmx13,5mm) 4x	22N, 45N
Hitrost preše	100RPM, 150RPM, 200RPM, 250RPM

Elektro komponente:

Uporovni termometer MWT180/A – pt100, 0.1°C resolucija	Temp. okolice
Uporovni termometer MWT180/A – pt100, 0.1°C resolucija	Temp. okolice
Uporovni termometer MWT180/A – pt100, 0.1°C resolucija	Temp. orodja
Uporovni termometer MWT180/A – pt100, 0.1°C resolucija	Temp. ohišja aktuatorja
Termistor, PTC 1000	Egasi – temp. navitja
Kollmorgen AKD 6A	Servo ojačevalnik
Siemens S7 1215 – PLK	Procesni krmilnik

Elektro-mehanske komponente:

Egasi - servo aktuator	Translacijski aktuator
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Oprema:

Stiskalnica	RVD-32-720HS
Orodje	89990-11
Izdelek (Vtiskovanje v trak) – testiranje brez traku	39900-51 (#40461 – DC01 0,50x30) Pomik 16mm Hod: 20mm Hitrost: 250min-1

Programske nastavitve:

Kollmorgen AKD, servo – hitrost	250mm/s
Kollmorgen AKD – pospešek	5600mm/s ²
Kollmorgen AKD – pojemek	5600mm/s ²
Kollmorgen AKD – max. tok	1.5Arms
Siemens – Aktivacijski kot preše	270° - 90°
Ciklični čas PLK	5ms



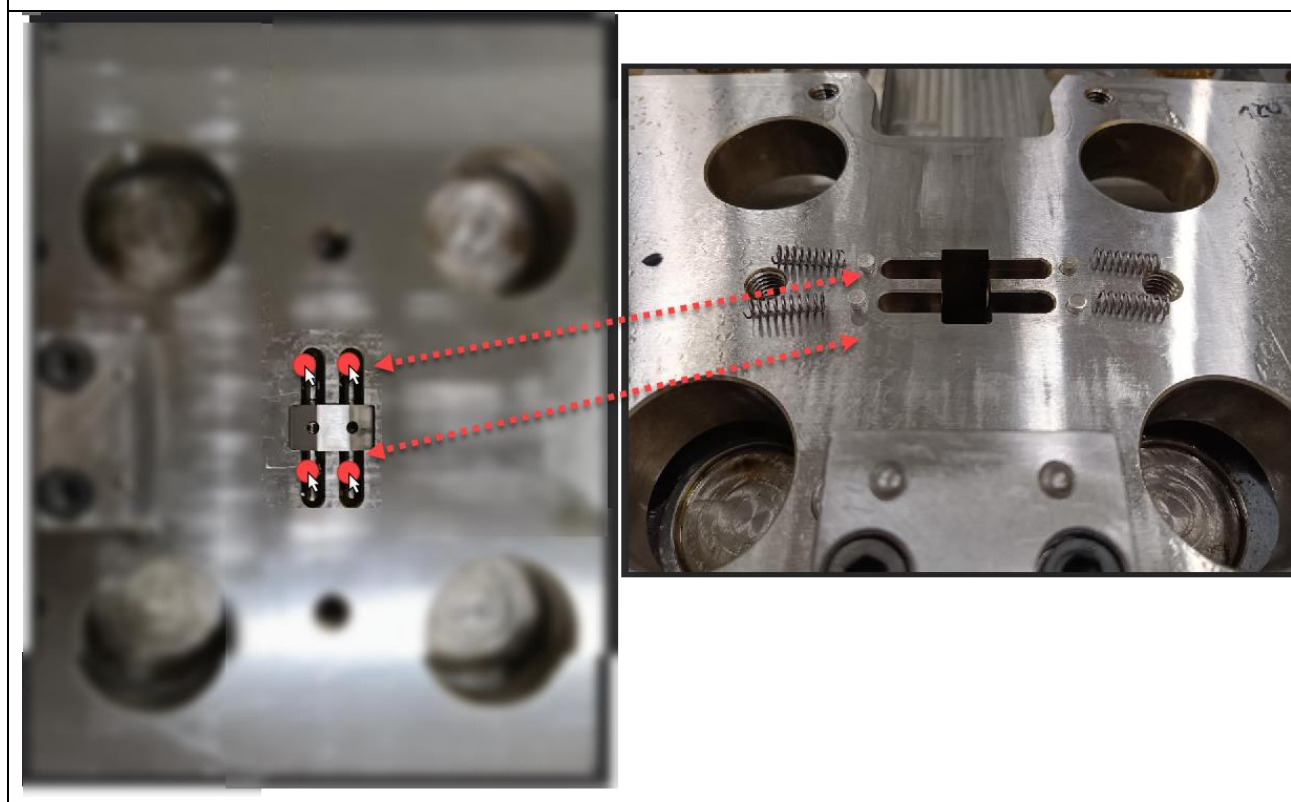
Rezultati

Vertikalna sila (N)	Hitrost (min-1)	Max. tok (A)	Min tok (A)	Efektivni tok I_{rms} (A)	Temperatura Egasi (°C)
22	100	2,313	-1,76	0,463	30
22	150	2,419	-1,835	0,580	32,5
22	200	2,423	-1,796	0,649	35
22	250	2,404	-1,743	0,712	35
45	100	4,261	-2,835	0,774	30
45	150	4,183	-2,554	0,888	35
45	200	4,131	-2,513	0,994	35
45	250	4,146	-2,363	1,048	40

Podložen pestič – vzmet 0,55mmx4mmx13,5mm - **dosežena vertikalna sila 22N**

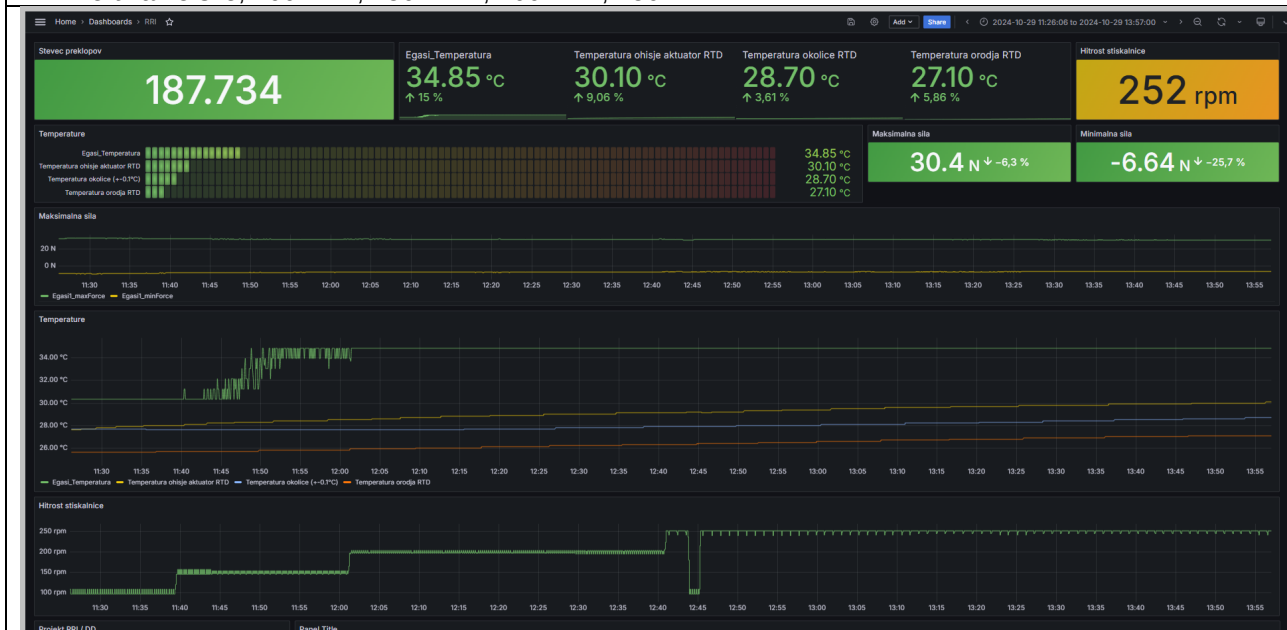
Podložen pestič – vzmet 0,55mmx4mmx13,5mm + 2mm podloga – **dosežena vertikalna sila 45N**

Referenčna vzmet poslana v katedralo za izdelavne tehnologije in sisteme FS UL.

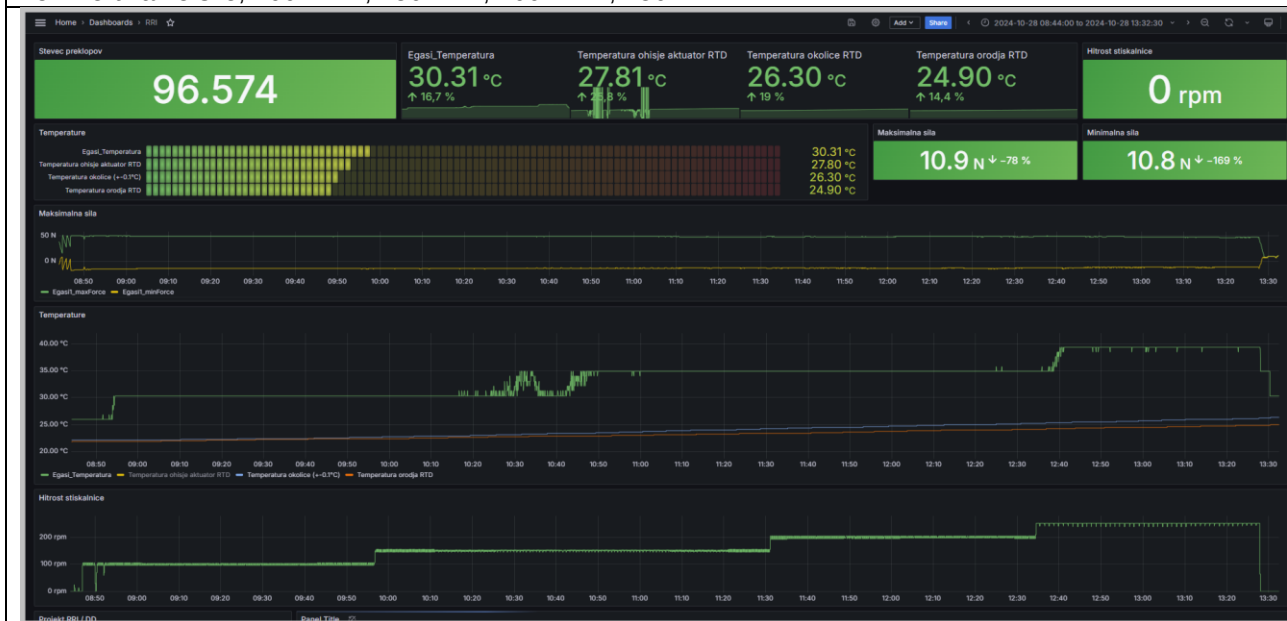




22N vertikalne sile, 100RPM, 150 RPM, 200RPM, 250 RPM

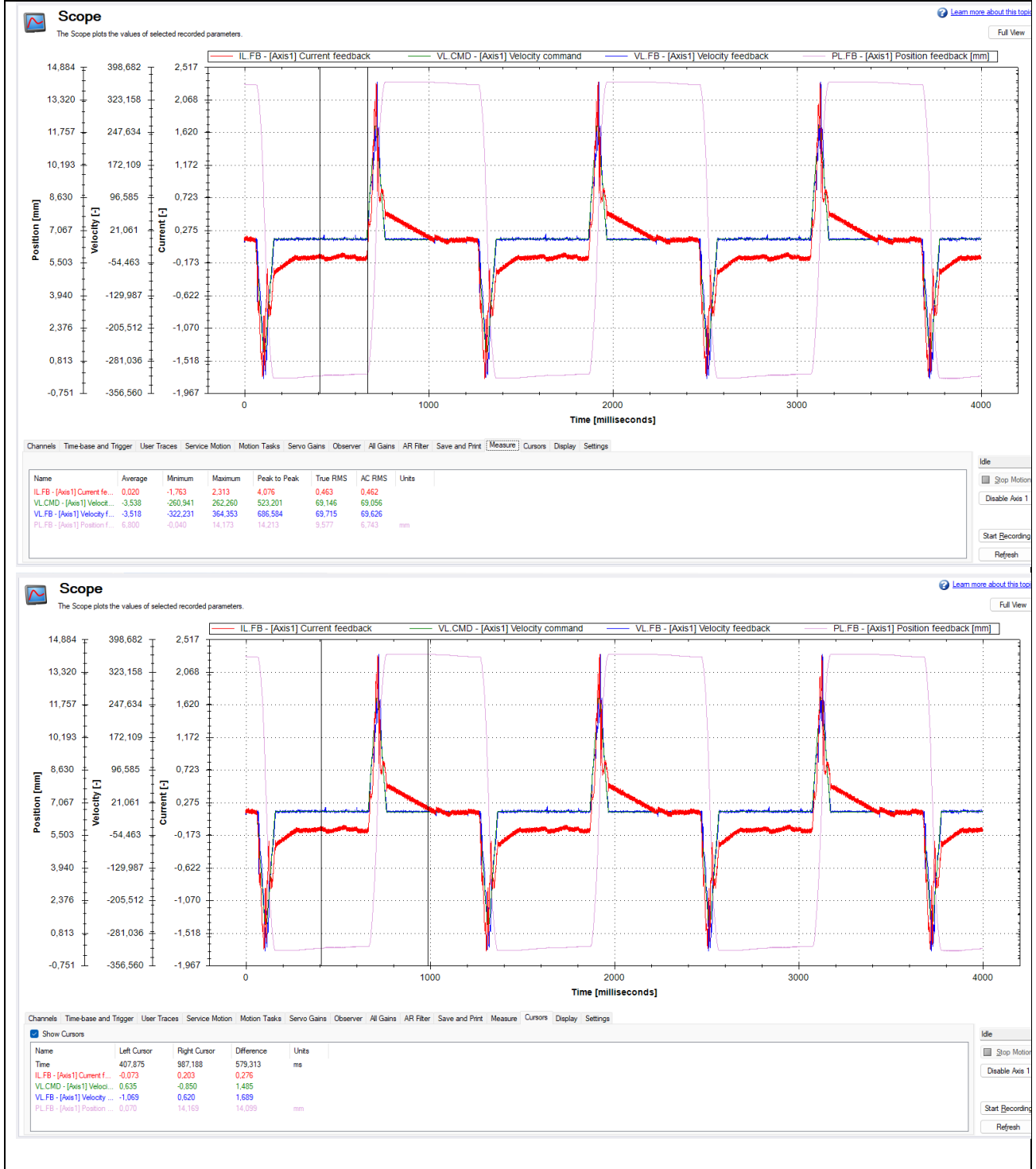


45N vertikalne sile, 100 RPM, 150RPM, 200 RPM, 250 RPM





22N vertikalne sile, 100 RPM





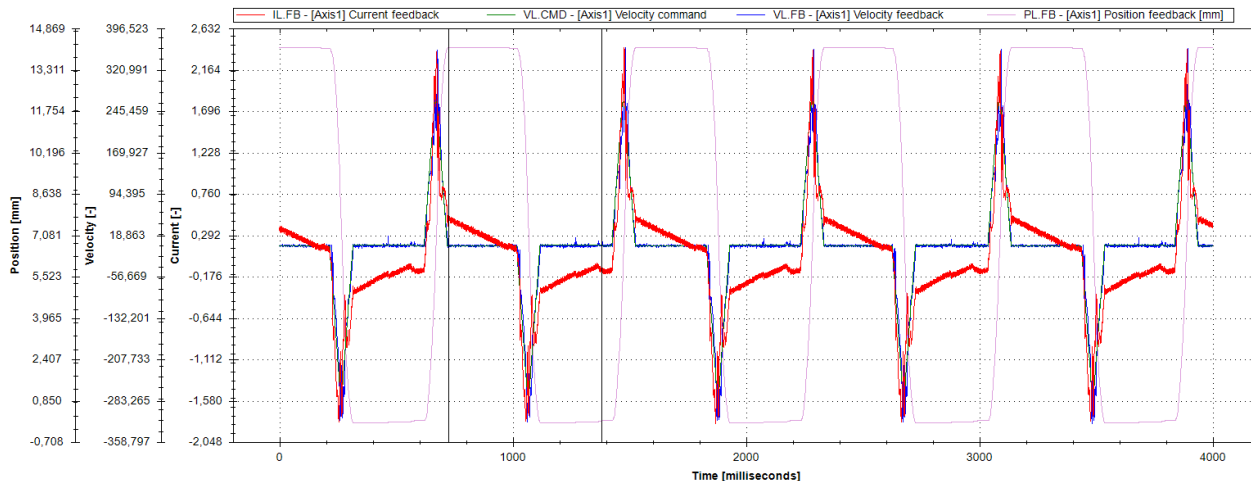
22N vertikalne sile, 150 RPM

Scope

The Scope plots the values of selected recorded parameters.

[Learn more about this tool](#)

Full View



Channels Time-base and Trigger User Traces Service Motion Motion Tasks Servo Gains Observer All Gains AR Filter Save and Print Measure Cursors Display Settings

Name	Average	Minimum	Maximum	Peak to Peak	True RMS	AC RMS	Units
IL FB - [Axis1] Current f...	0.050	-1.535	2.419	4.254	0.590	0.579	
VL CMD - [Axis1] Velocity...	0.004	-261.674	262.753	524.427	82.848	82.848	
VL FB - [Axis1] Velocity f...	-0.020	-324.464	362.190	686.654	83.156	83.156	
PL FB - [Axis1] Position f...	7.011	0.021	14.161	14.140	9.653	6.635	mm

Idle

Stop Motion

Disable Axis 1

Start Recording

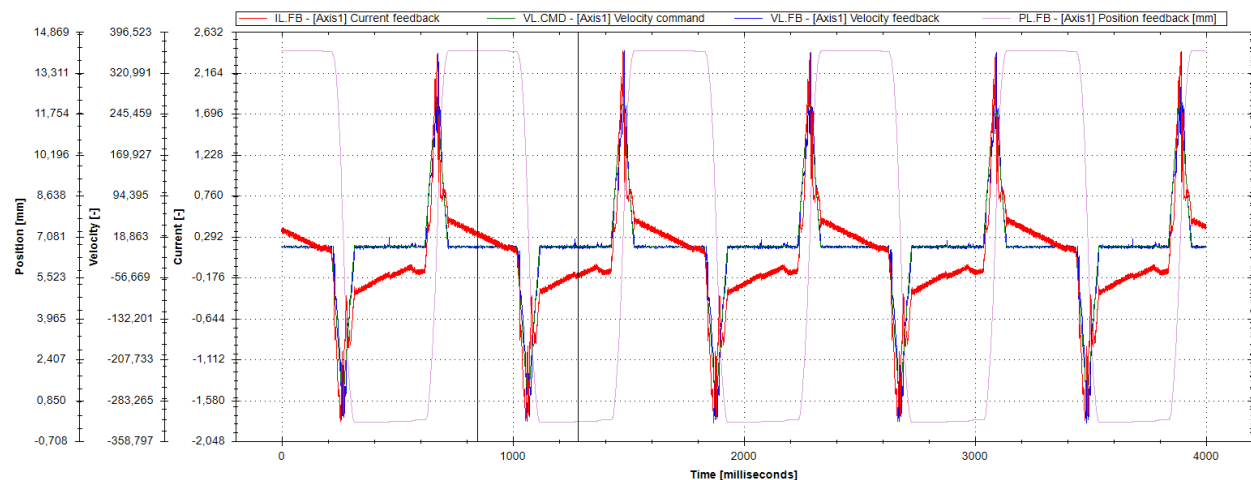
Refresh

Scope

The Scope plots the values of selected recorded parameters.

[Learn more about this tool](#)

Full View



Channels Time-base and Trigger User Traces Service Motion Motion Tasks Servo Gains Observer All Gains AR Filter Save and Print Measure Cursors Display Settings

Name	Left Cursor	Right Cursor	Difference	Units
Time	846.625	1.282.125	435.500	ms
IL FB - [Axis1] Current f...	0.337	-0.175	0.512	
VL CMD - [Axis1] Velocity...	-0.762	0.754	1.516	
VL FB - [Axis1] Velocity f...	-1.344	1.772	3.116	
PL FB - [Axis1] Position ...	14.152	0.051	14.101	mm

Idle

Stop Motion

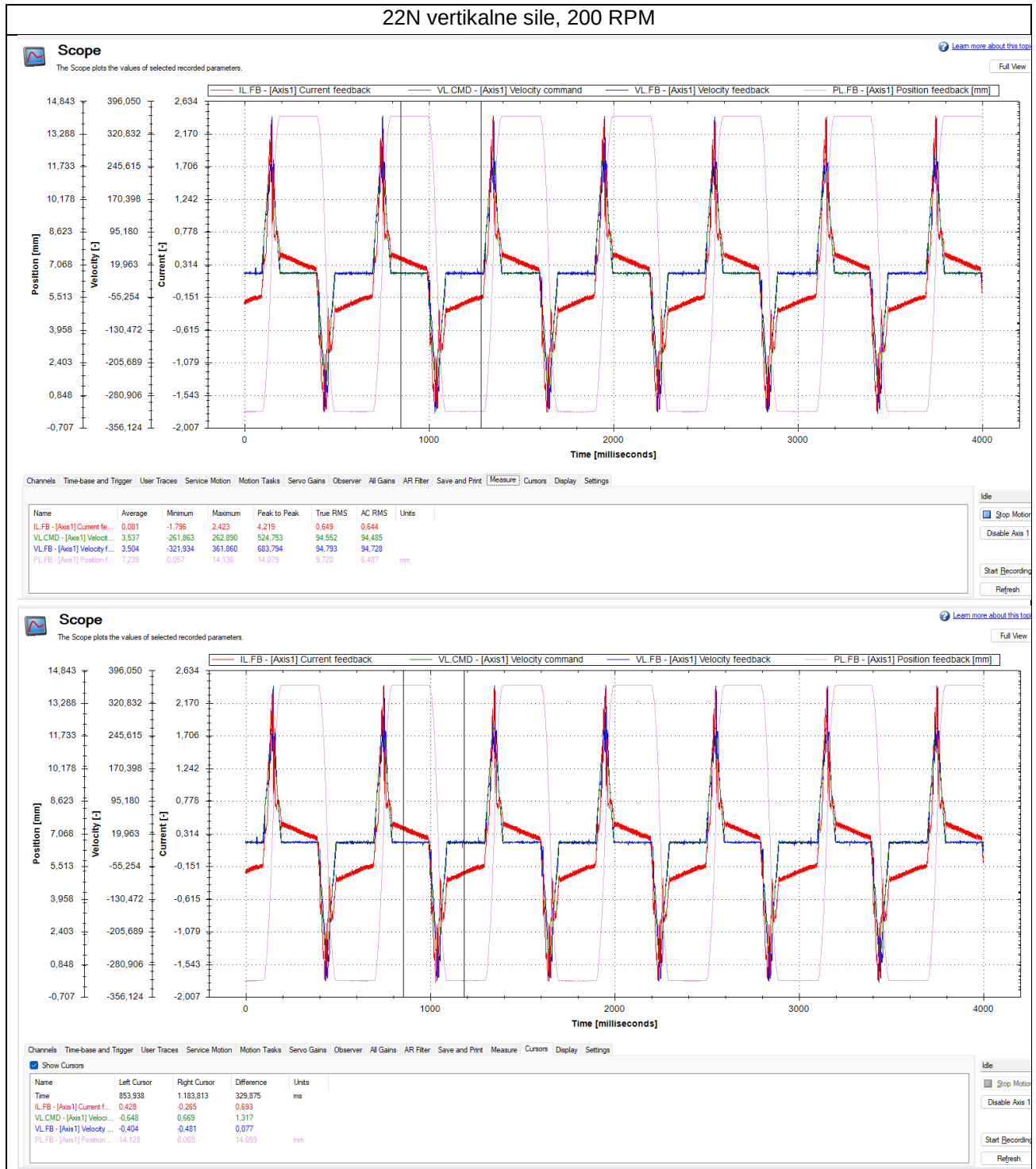
Disable Axis 1

Start Recording

Refresh

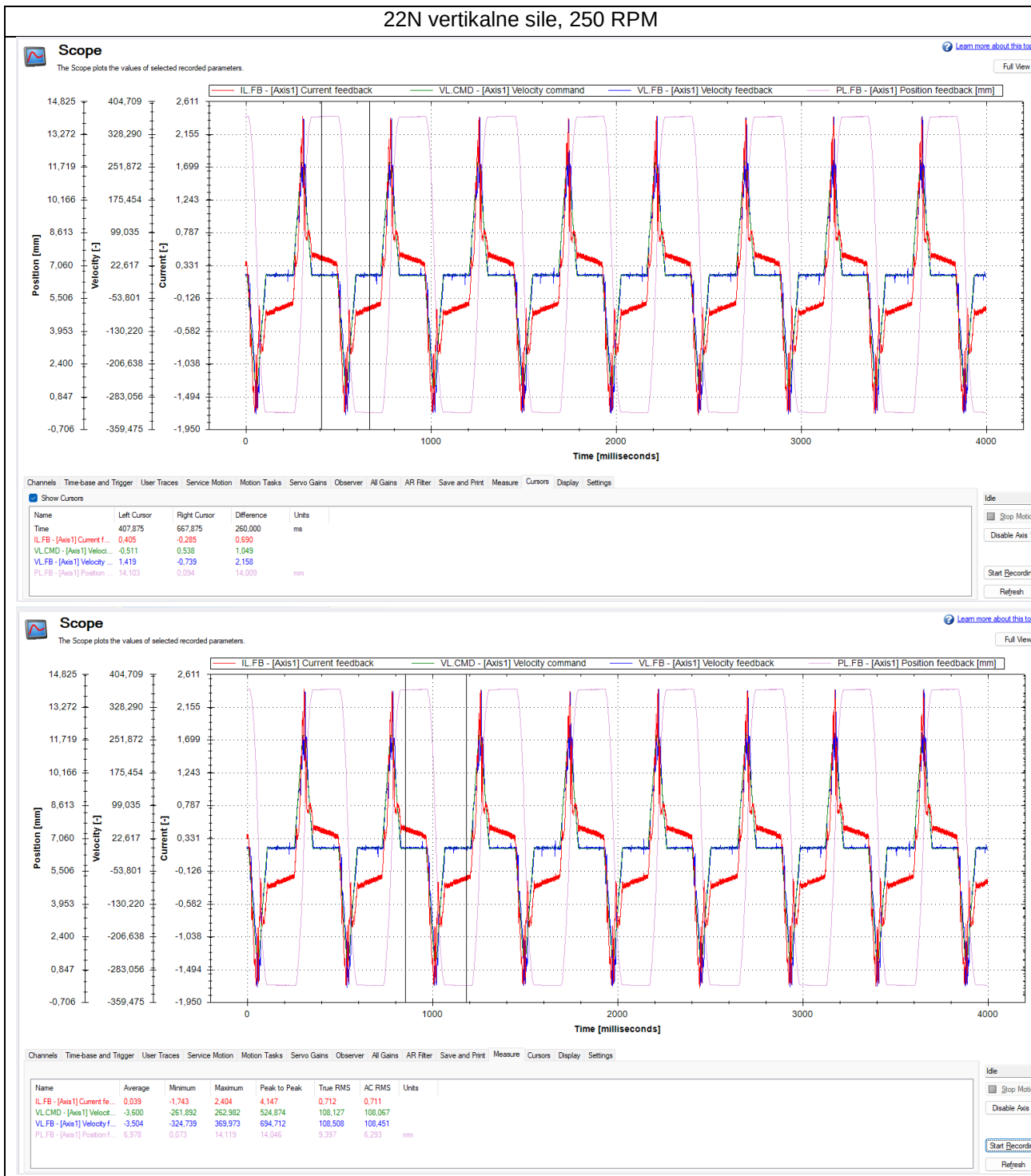


22N vertikalne sile, 200 RPM



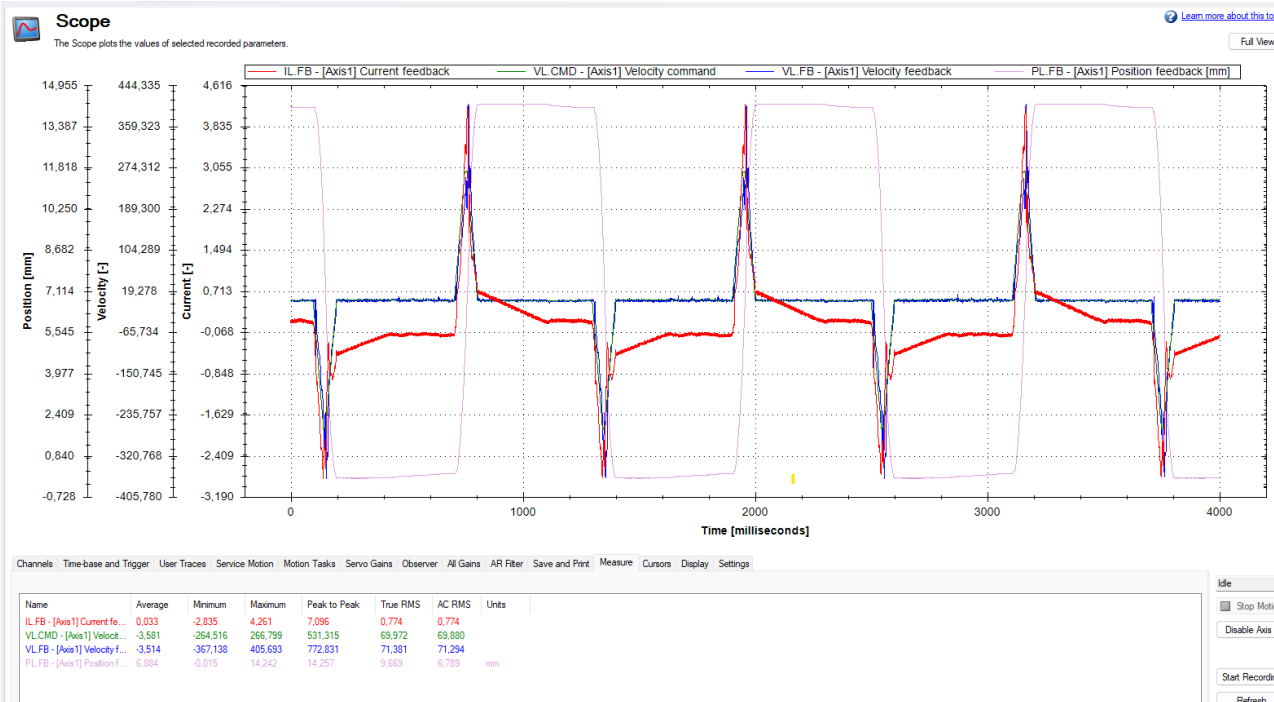
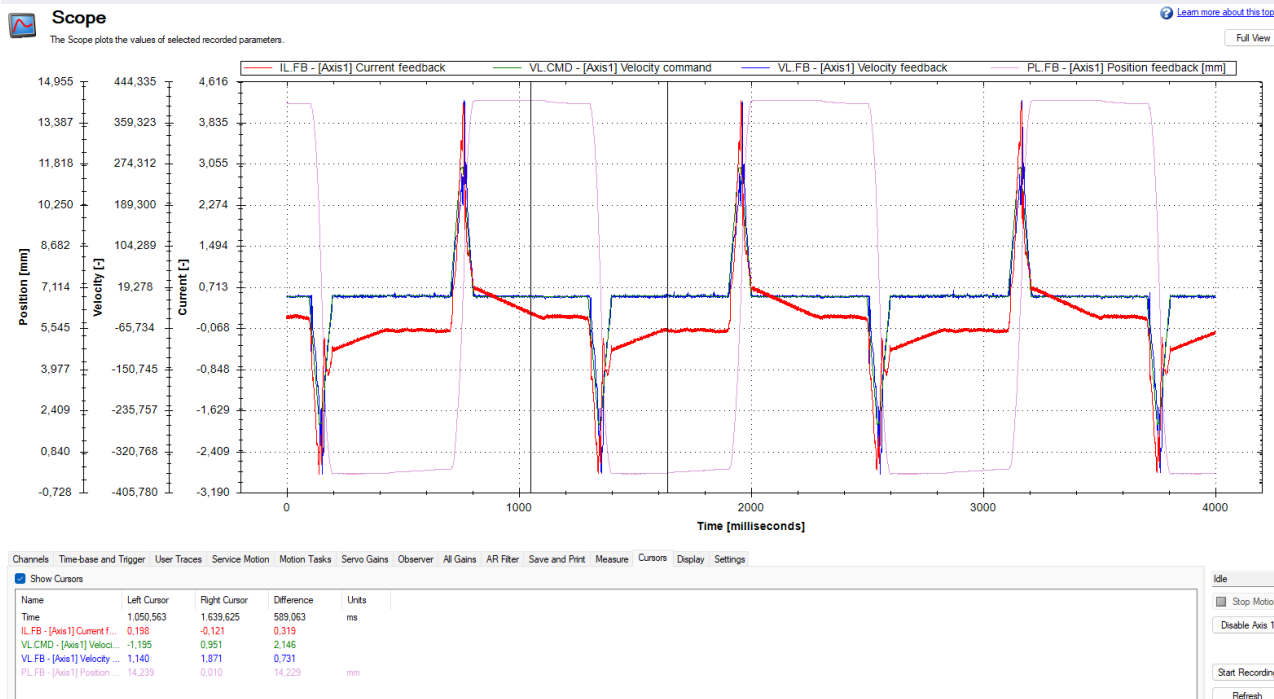


22N vertikalne sile, 250 RPM



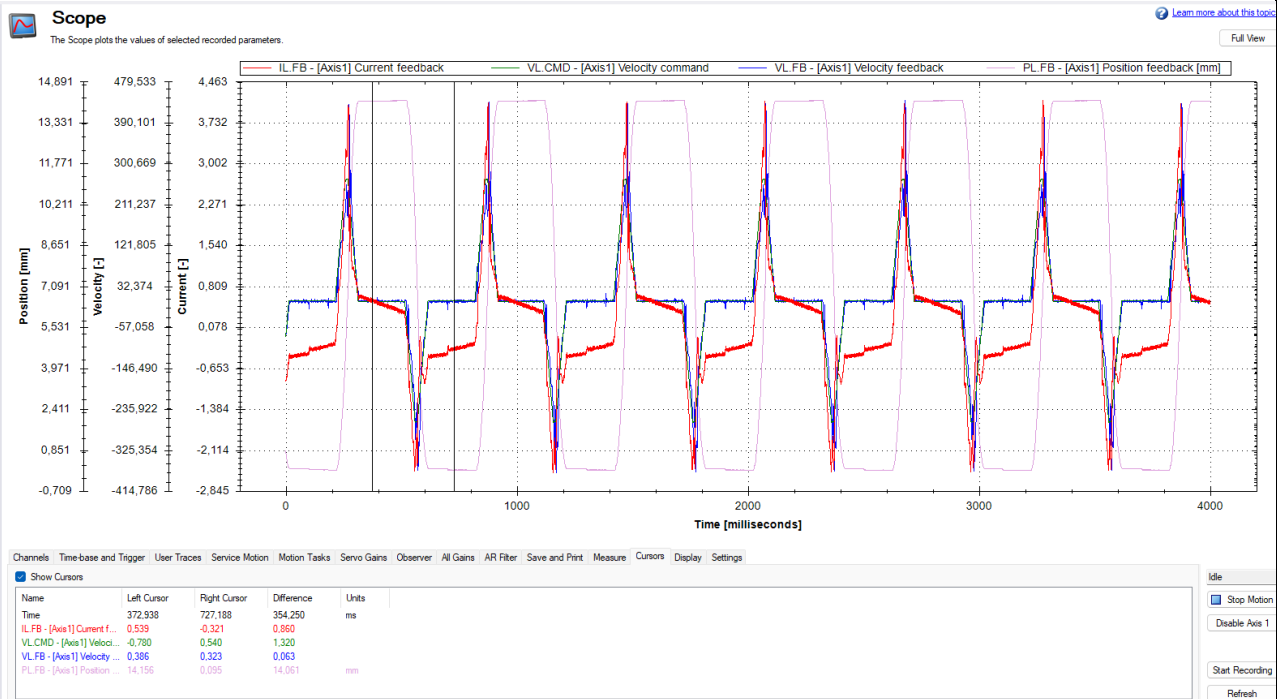
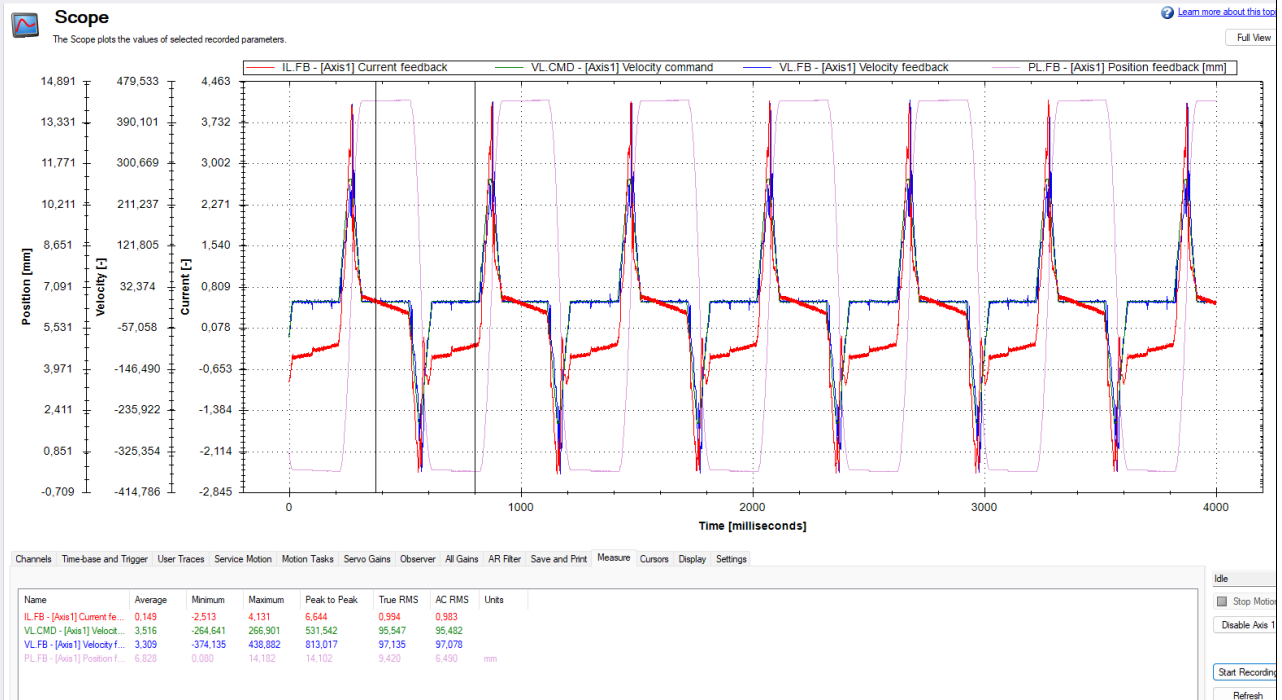


45N vertikalne sile, 100 RPM



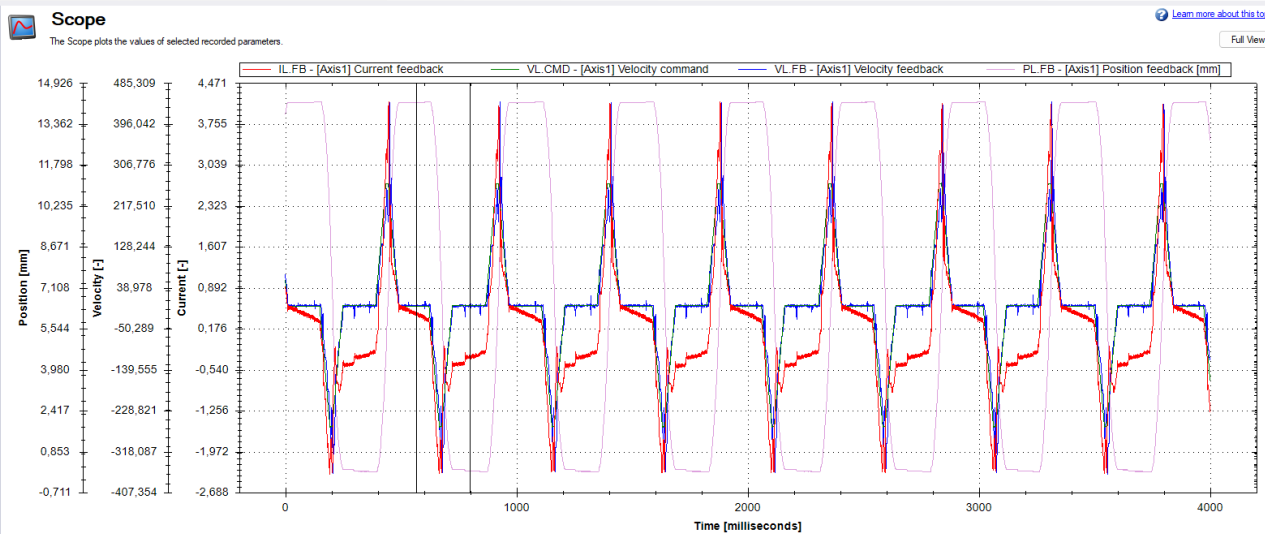
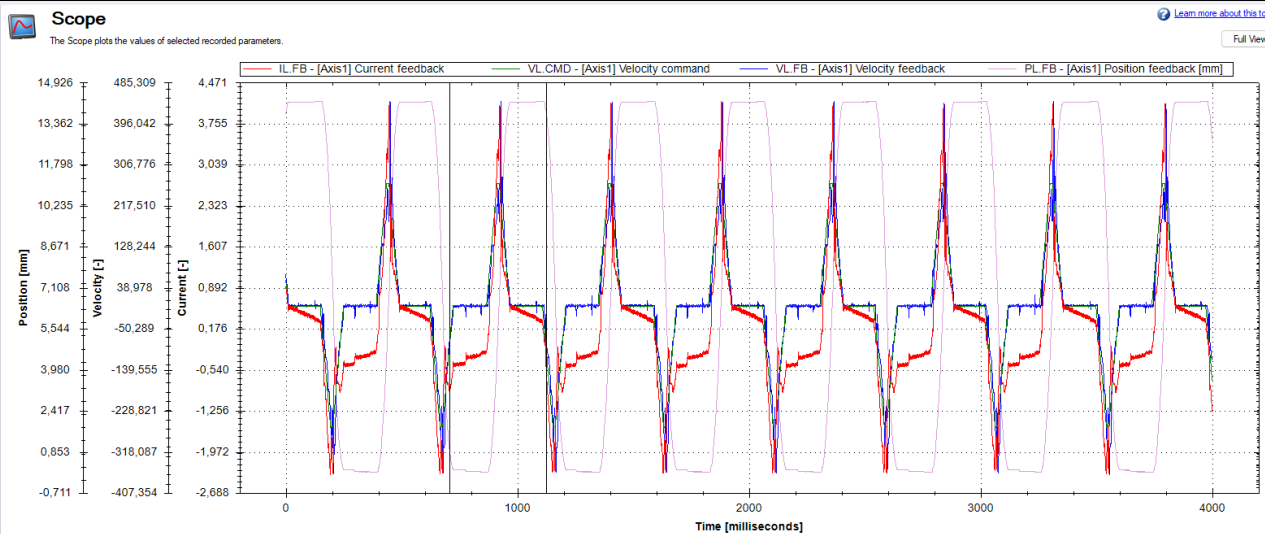


45N vertikalne sile, 200 RPM

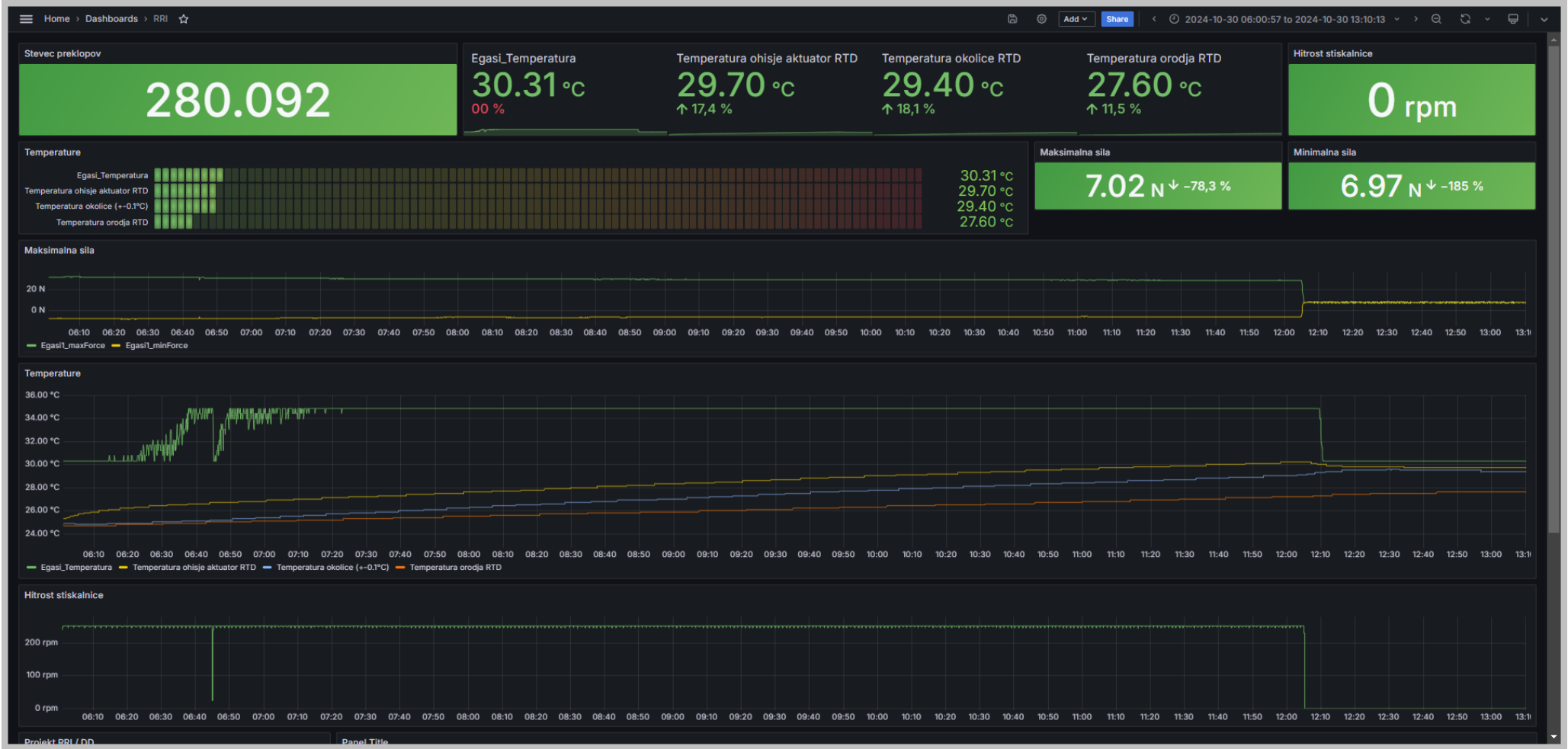




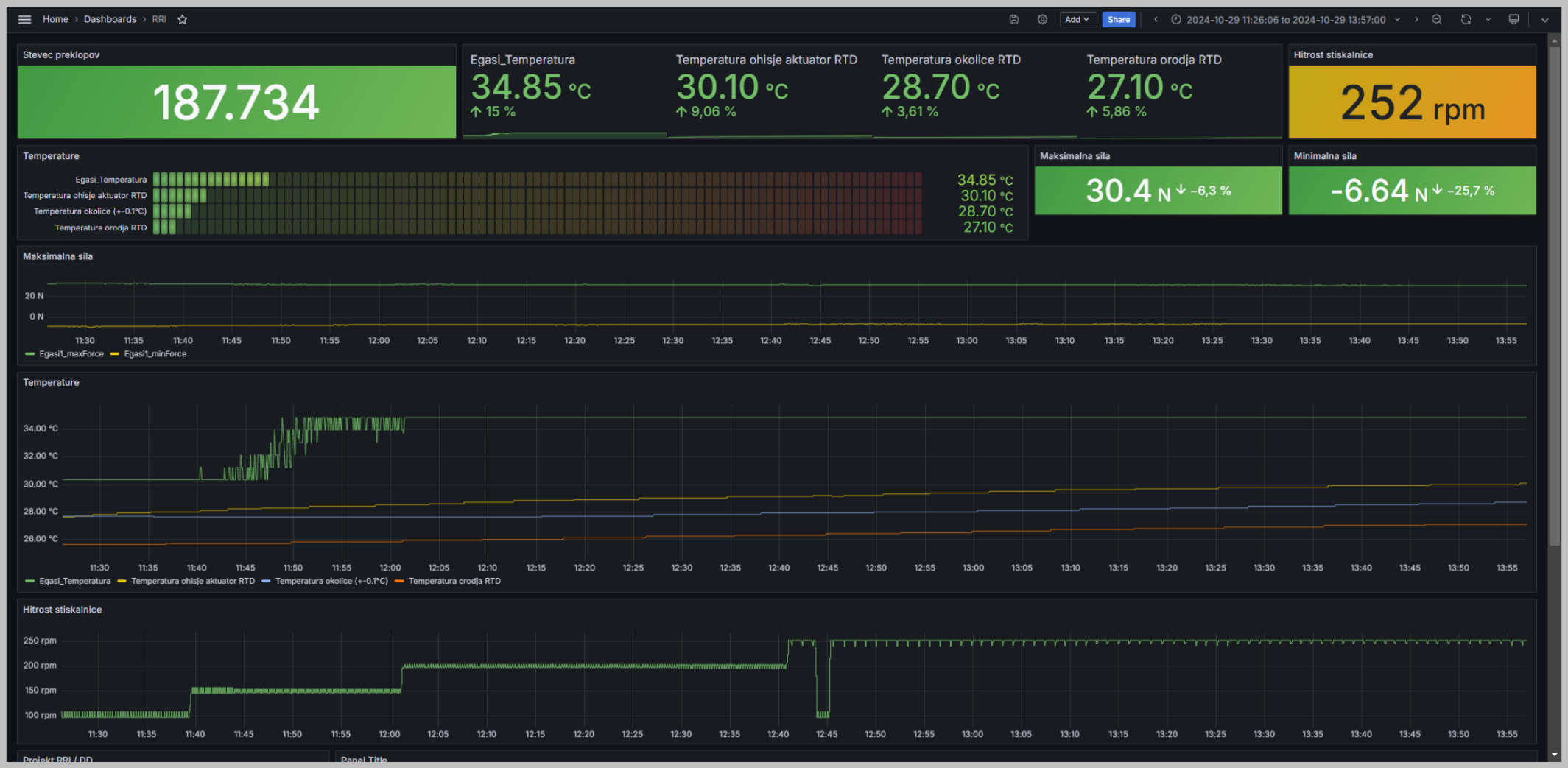
45N vertikalne sile, 200 RPM



22N vertikalne sile, 250 RPM (6h)



22N vertikalne sile | @100RPM, @150RPM, @200RPM, @250RPM



45N vertikalne sile | @100RPM, @150RPM, @200RPM, @250RPM

